"PAPER_{0:D3}" -f [int]# PAPER #78 ■ Extragalactic Physics: NED Multi-Wavelength + UQFF

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<!-- UQFF constants: ? = 5.0e-4 day?■, [SSq] = 0.57, M_UQFF = 1.43e1 TeV -->

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1. UQFF Hubble Constant Analysis

Hubble Tension Context

Measurement	H0 (km/s/Mpc)	Method
Planck 2018 (CMB)	67.4 ■ 0.5	Early universe
SH0ES 2023 (Cepheids)	73.0 ■ 1.0	Distance ladder
Tension	4.2s	■

UQFF Buoyant Correction to H0

The UQFF vacuum buoyancy modifies the effective expansion rate:

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H_{\rm UQFF}(z) = H_0 \times \sqrt{\Omega_\Lambda + \Omega_m(1+z)^3 + [UA] \times
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The [UA] = 0.0001 fractional vacuum coupling adds:

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\Delta H_0 = H_0 \times [UA] \times 0.5 = 67.4 \times 0.0001 \times 0.5 = 0.0034 \text{
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Redshift bin	L*_standard (L?)	L*_UQFF (L?)	NED data range
z = 0.5	$10^{45.0}$	$10^{45.3}$	$10^{44.8}$■$10^{45.5}$
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Observable	NED Data	UQFF Prediction	Agreement
H0 tension	5.6 km/s/Mpc gap	?H0 = 0.003 (negligible)	Not resolved
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km/s/Mpc}
```

UQFF correction to Hubble tension: ?H0 = 0.003 km/s/Mpc ■ far too small to resolve the 5.6 km/s/Mpc tension. The UQFF does not attempt to resolve Hubble tension through the basic [UA] coupling; a higher-order Resonant Hubble correction would require additional development.

---

2. AGN Luminosity Function Comparison

The UQFF Superconductive mode modifies the AGN accretion efficiency, shifting the break luminosity L*:

```
L_*^{\rm UQFF} = L_*^{\rm standard} \times (1 + [SCm]) = L_* \times 1.99
```

NED quasar catalog comparison:

Redshift bin	L*_standard (L?)	L*_UQFF (L?)	NED data range
z = 0.5	$10^{45.0}$	$10^{45.3}$	$10^{44.8}$■$10^{45.5}$
z = 1.0	$10^{45.5}$	$10^{45.8}$	$10^{45.2}$■$10^{46.0}$
z = 2.0	$10^{45.8}$	$10^{46.1}$	$10^{45.5}$■$10^{46.3}$
z = 3.0	$10^{46.0}$	$10^{46.3}$	$10^{45.7}$■$10^{46.5}$

UQFF L* shift of 0.3 dex lies within the 0.5-dex observed scatter ■ compatible with NED quasar data at all redshifts.

---

3. Quasar Absorption Line Systems

DLA (Damped Lyman-a) systems contain high column density neutral hydrogen (N_HI > 2■10■■ cm?■). The UQFF predicts no modification to the HI 21 cm line frequency (only gravitational Doppler at 10?■■ level). NED DLA catalog: UQFF-consistent.

Summary

Observable	NED Data	UQFF Prediction	Agreement
H0 tension	5.6 km/s/Mpc gap	?H0 = 0.003 (negligible)	Not resolved
AGN L*	10^{45}■$10^{46.5}$	+0.3 dex ([SCm])	Within scatter
DLA HI column	N_HI > 10^{20}	Unmodified	Compatible

Source: QCalcvalidation.py NEDBASE endpoint | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value "PAPER$_{0:D3}$" -f $n

---

Abstract

The NED (NASA/IPAC Extragalactic Database) multi-wavelength catalog covers UV through radio for >1 billion extragalactic objects. Key physics tests for UQFF: (1) AGN luminosity functions comparing UQFF-enhanced accretion vs standard models, (2) the Hubble tension (H0 = 67■73 km/s/Mpc) examined through the UQFF Buoyant vacuum correction, and (3) quasar absorption line systems (DLA/LLS) testing the UQFF vacuum density at cosmological redshifts.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

1. UQFF Hubble Constant Analysis

Hubble Tension Context

Measurement	H0 (km/s/Mpc)	Method
Planck 2018 (CMB)	67.4 ■ 0.5	Early universe

Measurement	H0 (km/s/Mpc)	Method
SH0ES 2023 (Cepheids)	73.0 ■ 1.0	Distance ladder
Tension	4.2s	■

UQFF Buoyant Correction to H0

The UQFF vacuum buoyancy modifies the effective expansion rate:

```
H_{\rm UQFF}(z) = H_0 \times \sqrt{\Omega_\Lambda + \Omega_m(1+z)^3 + [UA] \times
\rho_{\rm vac,(UQFF)} \times 8\pi G / 3H_0^2}
```

The [UA] = 0.0001 fractional vacuum coupling adds:

```
\Delta H_0 = H_0 \times [UA] \times 0.5 = 67.4 \times 0.0001 \times 0.5 = 0.0034 \text{
km/s/Mpc}
```

UQFF correction to Hubble tension: ?H0 = 0.003 km/s/Mpc ■ far too small to resolve the 5.6 km/s/Mpc tension. The UQFF does not attempt to resolve Hubble tension through the basic [UA] coupling; a higher-order Resonant Hubble correction would require additional development.

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Appendix: UQFF Production Framework Reference (v4.75+)

> *Added by upgradeearlywhitepapers.py (v4.75). This appendix cross-references
> the production physics constants and master equations to enable reproducibility
> against the current codebase state.*

A.1 Calibration Constants

Symbol	Value	Description
κ	$5.0 \times 10^{-■}$ day^{-1}	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60–0.61	Buoyancy coupling coefficient
k■	1.5	Ug1 DPM-dipole coupling
k■	1.2	Ug2 outer-bubble charge coupling
k■	1.8	Ug3 string-rotation coupling
k■	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
E_react(0)	10■■ J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

```
F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^{4}■igl[\lambda_i
\cdot U_i(r,t) \cdot E_{\mathrm{react}}■igr]
```

Term	Description	Implementation
Ug1	DPM magnetic dipole	`compute_Ug1_SOURCE4` / `compute_Ug1()`
Ug2	Outer-field bubble (charge-reactivity)	`compute_Ug2_SOURCE4` / `compute_Ug2()`
Ug3	Magnetic string rotation	`compute_Ug3_SOURCE4` / `compute_Ug3()`
Ug4	Vacuum concentration (star-BH)	`compute_Ug4_SOURCE4` / `compute_Ug4()`
Ubi	Buoyancy force	`compute_Ubi_SOURCE4` / `compute_Ubi()`
Um	Universal Magnetism (Heaviside-amplified)	`compute_Um_SOURCE4` / `compute_Um()`
–Σλ■·U■·E_react	4th dissipation term (PAPER_420)	`compute_FU_SOURCE4` / full pipeline

4th dissipation term parameters (PAPER_420): λ■=10^{-10}, λ■=10^{-12}, λ■=10^{-11}, λ■=10^{-13} (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

```
U_m^{\mathrm{full}} = U_m^{\mathrm{base}} imes ■igl(1 + 10^{13}\,\Theta(
ho_{SCm} -
```

```
ho_c)■igr) imes ■igl(1 + A_q\cos(\Delta\omega\,t)■igr)
```

Symbol	Value	Description
ρ_c	$10^{1■}$ kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
Δω	2π/(434·365.25) rad/day	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	β_i × Ubi	Expanding nebulae, stellar winds
Superconductive	Um × (1+10^{13}·f_H)	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in `MAIN_1_CoAnQi.cpp`, `CondensedPhysics.py`, *and* `CondensedPhysics2.py`.